

ULLARED, Sweden

WMO: 025390

Lat: 57.1134N

Lon: 12.7732E

Elev: 122

StdP: 99.87

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-11.7	-16.5	0.9	-14.0	-13.9	1.1	-9.6	5.5	1.1	5.0	1.2	0.3	40	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	26.6	18.1	24.8	17.4	22.9	16.6	19.6	24.1	18.6	22.9	17.6	21.4	1.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.2	21.1	17.1	12.4	20.0	16.1	11.6	19.2	56.3	24.2	53.1	22.9	50.1	21.6	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.4	3.9	3.4	DB	-17.7	28.9	3.6	2.1	-20.3	30.4	-22.5	31.7	-24.5	32.9	-27.2	34.4	(4)
(5)			WB	-18.3	21.7	3.6	1.2	-20.9	22.5	-23.0	23.3	-25.0	24.0	-27.6	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-2.0	-1.2	1.5	5.8	10.4	13.6	16.0	15.3	11.7	7.1	3.4	0.5		
	DBStd	7.18	4.61	4.03	3.76	3.37	3.86	2.94	2.74	2.52	2.80	3.65	3.78	4.86		
	HDD10.0	1734	372	313	262	131	41	3	0	1	16	103	198	294		
	HDD18.3	4197	630	546	521	375	248	144	84	99	201	348	447	553		
	CDD10.0	600	0	0	0	6	53	112	186	164	66	13	1	0		
	CDD18.3	21	0	0	0	0	2	3	12	4	0	0	0	0		
	CDH23.3	337	0	0	0	1	29	82	172	50	3	0	0	0		
	CDH26.7	49	0	0	0	0	2	10	33	4	0	0	0	0		
(14) Wind	WSAvg	1.4	1.6	1.7	1.6	1.5	1.4	1.2	1.0	1.1	1.2	1.4	1.7	1.7		
(15) Precipitation	PrecAvg	1159	110	82	71	61	71	93	110	117	109	121	105	110		
	PrecMax	1456	215	161	174	129	111	163	245	198	258	260	177	249		
	PrecMin	815	3	24	25	9	13	0	20	18	25	34	51	16		
	PrecStd	184	59	43	35	33	25	39	53	48	57	56	39	63		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	8.2	14.6	20.6	26.2	28.0	29.8	26.9	23.0	17.2	12.1	9.5		
		MCWB	5.9	6.0	8.2	12.3	15.9	17.4	19.5	19.0	16.9	13.7	11.3	8.9		
	2%	DB	5.6	6.3	10.8	18.0	23.6	25.9	27.2	24.7	19.9	14.4	10.6	8.3		
		MCWB	5.1	5.4	6.8	10.3	15.1	17.3	18.8	18.4	15.3	12.9	9.9	7.8		
	5%	DB	4.8	5.2	8.1	15.4	21.2	23.3	25.3	22.7	18.1	13.4	9.7	7.4		
		MCWB	4.3	4.5	5.7	9.3	14.0	16.2	18.1	17.5	14.8	12.2	9.0	6.9		
	10%	DB	3.8	4.2	6.4	12.9	18.3	20.6	22.9	20.7	16.6	12.4	8.7	6.5		
		MCWB	3.4	3.6	4.7	8.1	12.9	14.9	17.3	16.8	14.0	11.4	8.0	6.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	6.6	8.7	13.2	18.2	19.6	21.4	20.6	18.2	14.4	11.5	9.1		
		MCDB	6.5	7.1	12.2	19.6	23.7	25.3	26.4	24.1	20.5	15.6	11.9	9.4		
	2%	WB	5.2	5.6	7.3	11.4	16.5	18.1	20.0	19.1	16.4	13.4	10.0	8.0		
		MCDB	5.5	6.0	9.7	15.9	21.0	23.9	24.9	22.7	18.9	14.2	10.4	8.3		
	5%	WB	4.4	4.5	6.1	10.0	14.9	16.8	19.0	18.2	15.2	12.4	9.1	6.9		
		MCDB	4.7	5.0	7.7	14.1	19.4	21.7	23.6	21.4	17.3	13.2	9.6	7.3		
	10%	WB	3.5	3.5	5.1	8.6	13.4	15.7	17.9	17.2	14.4	11.6	8.1	6.0		
		MCDB	3.8	4.1	6.3	11.9	17.3	19.7	21.9	19.9	16.2	12.3	8.6	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.5	7.9	11.0	11.6	11.1	10.7	9.7	9.1	7.2	4.9	4.5		
		MCDBR	4.3	5.3	10.8	16.1	16.0	16.6	15.7	13.4	11.6	7.6	4.9	4.0		
	5% WB	MCWBR	4.4	4.5	7.4	9.6	8.4	8.9	8.1	7.5	7.6	5.9	4.6	4.0		
		MCWBR	4.4	4.6	9.3	13.8	13.5	13.8	13.1	10.8	10.1	6.6	4.5	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.327	0.345	0.381	0.378	0.369	0.391	0.391	0.359	0.346	0.322	0.289		
		taud	2.268	2.314	2.346	2.304	2.378	2.428	2.381	2.386	2.460	2.406	2.320	2.223		
	Ebn at Noon		571	718	809	830	859	872	841	814	790	690	539	470		
		Edh at Noon	50	73	91	111	110	106	109	102	82	67	48	40		
(44) All-Sky Solar Radiation	RadAvg		0.39	1.00	2.38	3.90	5.10	5.45	5.02	3.99	2.79	1.31	0.49	0.27		
	RadStd		0.08	0.16	0.24	0.54	0.48	0.42	0.64	0.38	0.34	0.21	0.09	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	-126	-178	N/A	N/A

Nomenclature: See separate page